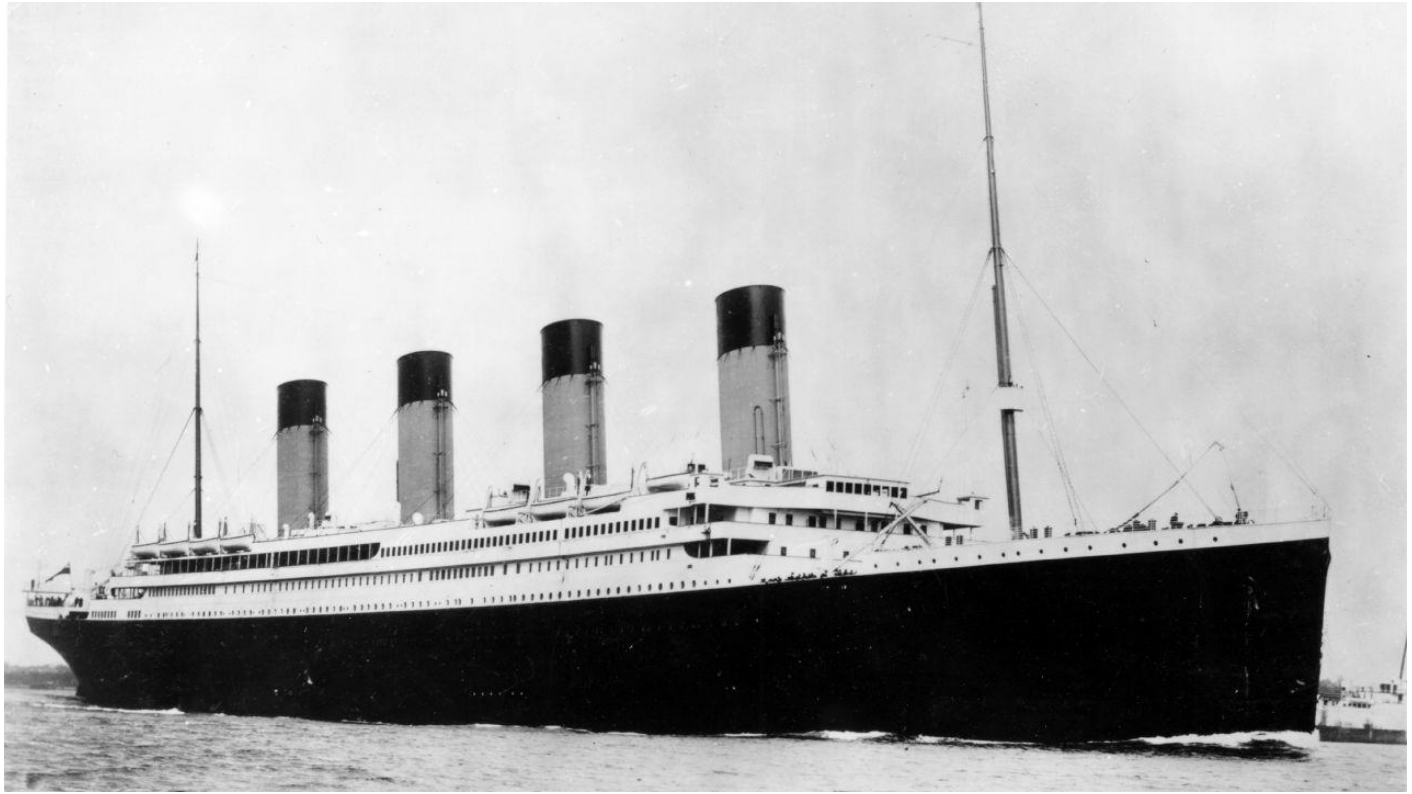
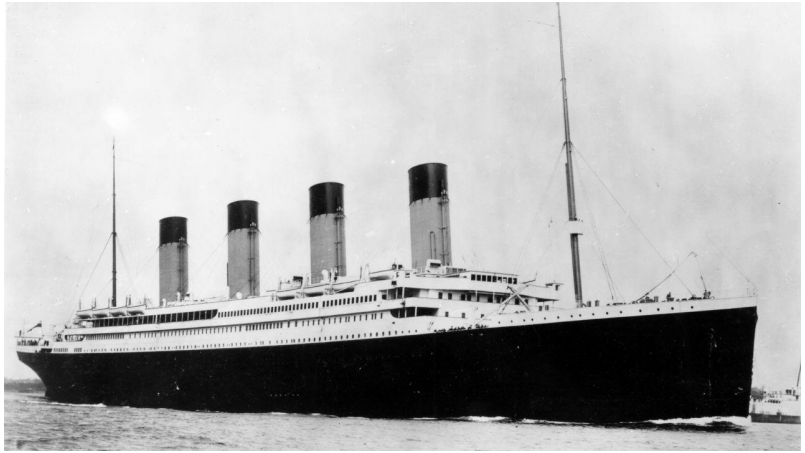
Watertight Bulkheads



What do you know about the Titanic?

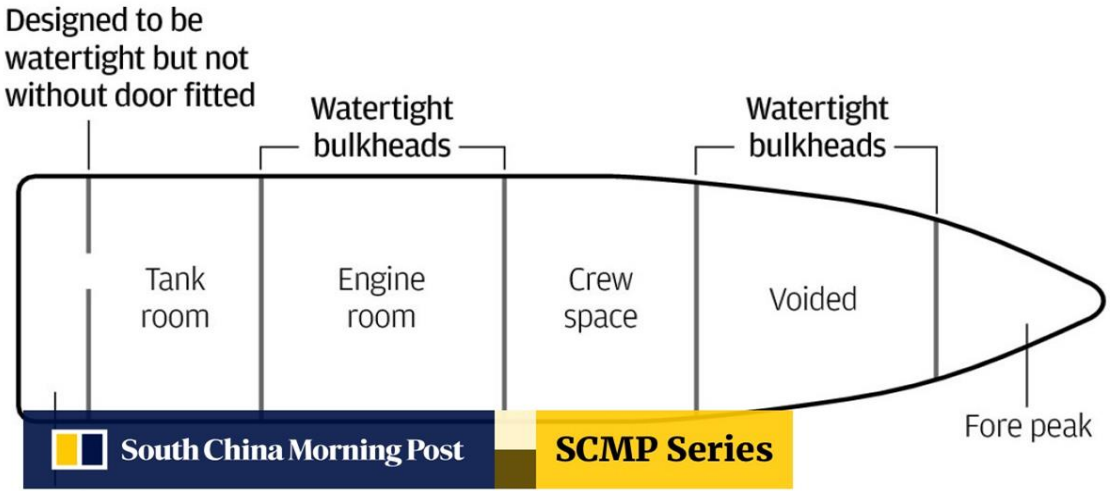


- Titanic weighed 46, 000 tonnes
- A ship will float providing it is able to displace a larger weight of water that its own weight
- When the Titanic struck the iceberg, a series of holes punctured the first five compartments, and as these filled up with water the bow of the ship tilted forward
- Once this was below the waterline there was nothing to stop the water replacing the air in other compartments, and since water is heavier than air, gravity became bigger than upthrust and the Titanic sank.

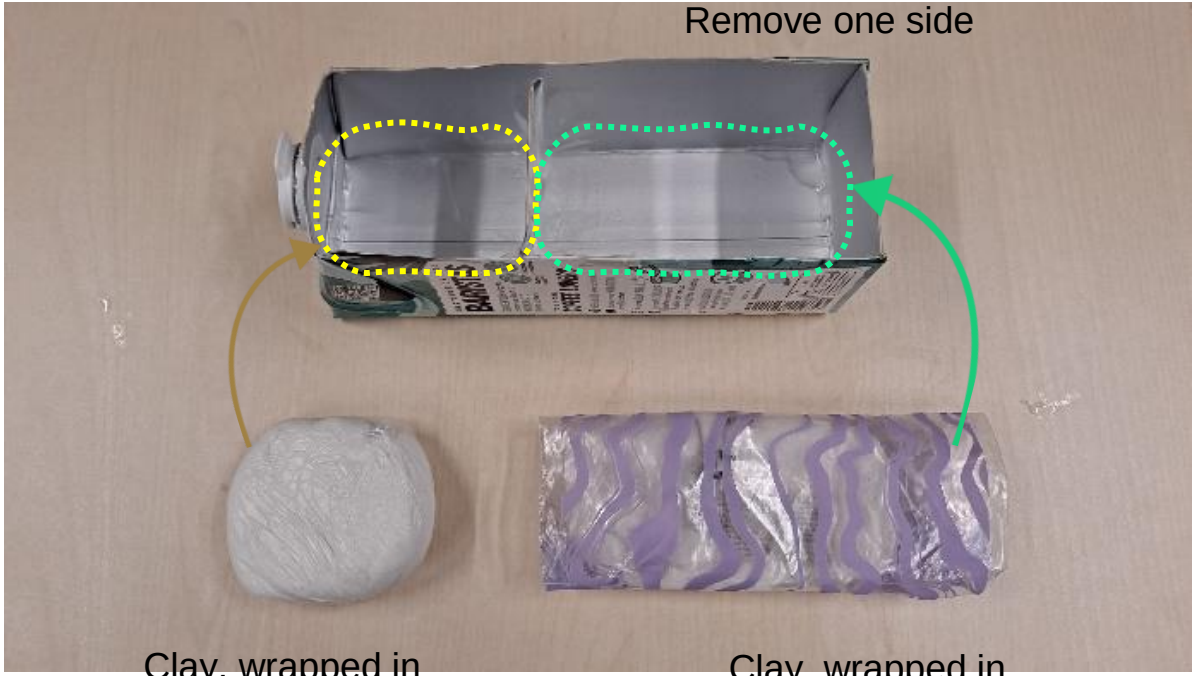


Watertight Bulkheads

Watertight bulkheads are vertically designed watertight divisions/walls within the ship's structure to avoid intake of water in the compartment if the adjacent compartment is flooded due to damage in ship's hull



Juice carton (waterproof)
Remove one side



Clay, wrapped in waterproof bags

Clay, wrapped in waterproof bags

Carton ship layout

- Build two identical model ships: one without a bulkhead, and one with a bulkhead.
- Place equal weights (clay in resealable bag) inside both ships to simulate the cargo and passengers. Ensure the weight is evenly distributed.
- Float the ships in their respective water tanks and observe if they remain stable and at the same water level.



With bulkhead



Without bulkhead

Method

- Using a pencil, poke a small hole at the same height (about 1.5 cm from the bottom) on both ships to simulate a breach in the hull.
- Observe and time how long it takes for each ship to begin to sink. Observe and record what happens in each ship. Repeat this process three times.



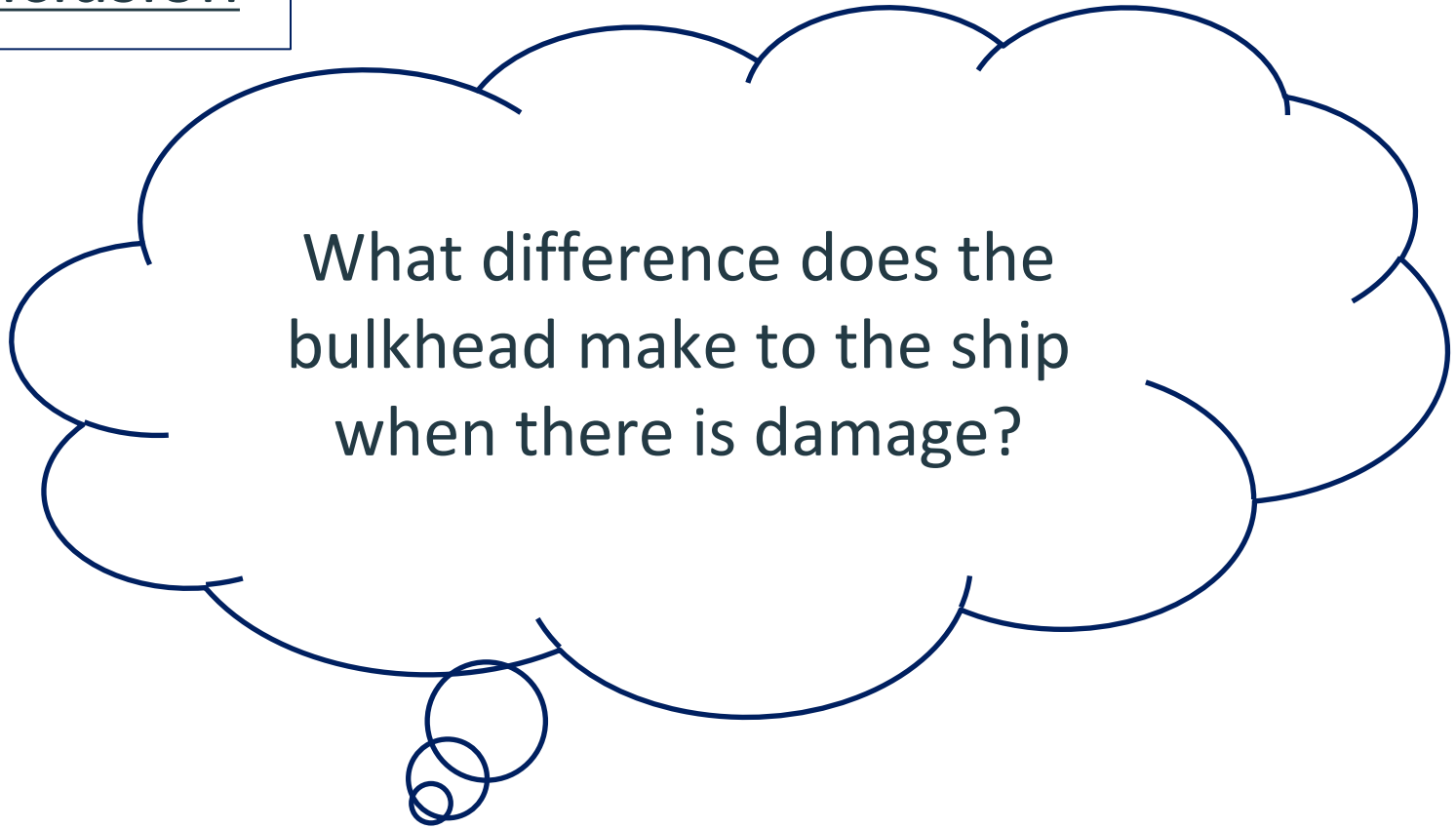
Variables

Independent variable - bulkhead

Dependent variable - time it takes to sink

Control variable – mass and distribution of clay, size of the hole, boat size …

Conclusion


What difference does the bulkhead make to the ship when there is damage?

The watertight bulkheads on the Titanic

Fifteen transverse bulkheads created sixteen compartments or cells, each of which could be isolated from the adjoining compartment using special doors which could be closed in the event of an emergency.

The lower section of the Titanic was divided into sixteen major watertight compartments that could easily be sealed off if part of the hull was punctured and leaking water.

When the hull of the Titanic was torn open in the collision with the iceberg, water began to flood the damaged compartments in the bow. As the ship pitched forward under the weight of the water in the bow compartments, water began to spill over the tops of the bulkheads into adjacent, undamaged compartments.

The iceberg breached 6 compartments. However, the damage to the 6th was fairly minor, so the crew were able to shore up. Unfortunately, 5 compartments flooding was enough to pull the tops of the bulkheads below the waterline, meaning they all eventually flooded.

remembering titanic

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